

# SKYPATH

Airline Booking Platform

## PROJECT REQUIREMENTS

UX, interaction, interface and accessibility requirements

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Design requirements and quality criteria

*This document translates the project's design objectives into practical requirements that can guide the structure, interaction behaviour and visual design of SkyPath.*

# 1. Purpose

This document defines the requirements that guide the design of the SkyPath Airline Booking Platform. It provides a reference point for evaluating whether proposed screens, flows and components support the intended user experience.

The requirements are written for a conceptual UX/UI project. They describe what the experience should support from a design perspective; they do not represent a production technical specification.

## 2. Requirement Principles

The requirements are based on five principles:

**Understandability:** users should be able to understand what information is being presented and what action is expected.

**Comparability:** important differences between flight and fare options should be easy to identify.

**Transparency:** significant costs, conditions and selections should be visible before commitment.

**Continuity:** information and decisions should remain coherent throughout the booking journey.

**Recovery:** users should receive clear feedback when an action succeeds, fails or requires correction.

## 3. Functional Experience Requirements

| ID    | Requirement                                                                                                                                                | Priority |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| FR-01 | The experience should allow users to begin a flight search by specifying core journey information such as origin, destination, travel date and passengers. | High     |
| FR-02 | Search results should present multiple flight options in a structure that supports direct comparison.                                                      | High     |
| FR-03 | Flight results should expose meaningful comparison attributes such as departure/arrival time, duration, stops and price.                                   | High     |
| FR-04 | Users should be able to inspect a selected flight before committing to a fare.                                                                             | High     |
| FR-05 | Fare options should communicate meaningful differences between available choices.                                                                          | High     |
| FR-06 | The booking journey should collect the passenger information required to continue the booking.                                                             | High     |
| FR-07 | Users should be able to select or review seat options where applicable.                                                                                    | Medium   |
| FR-08 | Optional services should be distinguishable from mandatory booking information.                                                                            | High     |
| FR-09 | Users should be able to review their selections before payment.                                                                                            | High     |
| FR-10 | The review stage should provide a clear price summary before confirmation.                                                                                 | High     |
| FR-11 | The experience should provide a clear confirmation state after a successful booking.                                                                       | High     |
| FR-12 | Post-booking access should allow the user to understand and revisit relevant trip information.                                                             | Medium   |

## 4. Information Requirements

The interface should present information according to its importance and timing within the journey. Information that affects a decision should be available before that decision becomes difficult to reverse.

| Area           | Information that should be clear                                                           |
|----------------|--------------------------------------------------------------------------------------------|
| Flight search  | Origin, destination, date, passenger count and journey type where applicable.              |
| Flight results | Airline, departure, arrival, duration, stops, price and relevant baggage/fare information. |

| Area              | Information that should be clear                                                           |
|-------------------|--------------------------------------------------------------------------------------------|
| Fare selection    | Fare name, included services, restrictions and meaningful differences between options.     |
| Passenger details | Required passenger fields, contact information and relevant validation feedback.           |
| Seat selection    | Seat availability/status, selected seat and any applicable seat-related cost or condition. |
| Optional services | What is being offered, whether it is optional, and its effect on the total.                |
| Review            | Selected flight, passengers, seats, services, price breakdown and final amount.            |
| Confirmation      | Booking reference, itinerary summary and clear next steps.                                 |

## 5. Interaction Requirements

| ID    | Requirement                                                                                                                    |
|-------|--------------------------------------------------------------------------------------------------------------------------------|
| IR-01 | Primary actions should be visually and semantically distinguishable from secondary actions.                                    |
| IR-02 | The interface should provide visible feedback after important user actions.                                                    |
| IR-03 | The current stage of the booking process should remain understandable through navigation, headings or progress cues.           |
| IR-04 | Users should be able to review important selections before final commitment.                                                   |
| IR-05 | Changing a selection should not unexpectedly erase unrelated information.                                                      |
| IR-06 | Interactive controls should have predictable states such as default, hover/focus, selected, disabled and error where relevant. |
| IR-07 | Destructive or consequential actions should require appropriate confirmation or provide a clear recovery path.                 |
| IR-08 | The experience should avoid unnecessary interaction steps when the same information can be communicated more efficiently.      |

## 6. Navigation Requirements

Navigation should support orientation without competing with the booking task. The primary journey should remain easy to follow, while users should retain reasonable access to previously entered or selected information.

| Requirement         | Design expectation                                                                                |
|---------------------|---------------------------------------------------------------------------------------------------|
| Orientation         | The current location or stage should be understandable.                                           |
| Consistency         | Navigation labels should use stable terminology throughout the experience.                        |
| Backtracking        | Users should be able to revisit appropriate previous stages without losing unrelated information. |
| Post-booking access | Completed trips should remain accessible after the booking journey.                               |

## 7. Form Requirements

Forms should minimise cognitive and visual burden while providing enough context for accurate completion.

**Field clarity:** labels should clearly communicate what information is required.

**Grouping:** related fields should be visually and conceptually grouped.

**Validation:** errors should identify the affected field and explain how the user can correct it.

**Progress:** long forms should be structured so users can understand how much information remains.

**Preservation:** avoid unnecessary loss of information when users navigate between stages.

## 8. Pricing and Transparency Requirements

Pricing is treated as a central trust factor in the booking experience.

**PR-01:** The price of a selected flight should be visible at meaningful decision points.

**PR-02:** Fare differences should be understandable without requiring users to infer what is included.

**PR-03:** Optional services should not appear indistinguishable from mandatory charges.

**PR-04:** The final review should communicate the total amount before payment.

**PR-05:** Important price changes caused by user selections should be reflected in the relevant summary.

## 9. Accessibility Requirements

| ID      | Requirement                                                                                                        |
|---------|--------------------------------------------------------------------------------------------------------------------|
| A11Y-01 | Text and interface elements should maintain sufficient contrast for comfortable reading and interaction.           |
| A11Y-02 | Information should not depend on colour alone to communicate status, selection or errors.                          |
| A11Y-03 | Interactive controls should have clear labels and understandable states.                                           |
| A11Y-04 | Keyboard/focus behaviour should be considered for interactive components in a production implementation.           |
| A11Y-05 | Typography and spacing should support readable content and clear hierarchy.                                        |
| A11Y-06 | Error messages should be specific and actionable rather than relying only on colour or visual emphasis.            |
| A11Y-07 | The design should consider users with different levels of digital confidence and familiarity with airline booking. |

## 10. Visual Design Requirements

| Area                | Requirement                                                                                                               |
|---------------------|---------------------------------------------------------------------------------------------------------------------------|
| Hierarchy           | The visual hierarchy should direct attention toward the most important decision or action on each screen.                 |
| Typography          | Type styles should establish a clear hierarchy between page titles, section headings, supporting information and actions. |
| Spacing             | Spacing should create grouping and separation without making the interface feel fragmented.                               |
| Components          | Repeated patterns should use consistent component structures and states.                                                  |
| Responsive thinking | Layouts should be designed with different viewport sizes and content lengths in mind.                                     |
| Brand expression    | Visual identity should support trust, clarity and travel context without reducing information legibility.                 |

## 11. Error Prevention and Recovery

The design should attempt to prevent avoidable mistakes before they occur and provide clear recovery when they do.

Examples include clear field labels, sensible grouping, visible selections, confirmation before consequential actions, meaningful validation messages and preservation of previously entered information.

## 12. Content and Language Requirements

Interface copy should be concise, direct and understandable. Labels should describe actions or information using familiar language rather than unnecessary technical terminology.

Important conditions, restrictions and costs should not be hidden behind vague labels. Where a user needs to make a decision, the interface should provide enough context to make that decision responsibly.

## 13. Component Requirements

| Component         | Expected states / considerations                                           |
|-------------------|----------------------------------------------------------------------------|
| Buttons           | Primary, secondary, disabled, focus and loading/processing where relevant. |
| Input fields      | Default, focus, filled, error and disabled states.                         |
| Flight cards      | Default, selected and comparison-relevant information.                     |
| Fare cards        | Default, selected and differentiated feature/inclusion states.             |
| Seat controls     | Available, selected, unavailable and contextual states.                    |
| Alerts / feedback | Success, warning, error and informational states where applicable.         |
| Navigation        | Default, active/current and responsive considerations.                     |

## 14. Prototype Requirements

The interactive prototype should represent the principal booking journey rather than attempting to prototype every possible edge case.

At minimum, the prototype direction should demonstrate the relationship between search, results, flight selection, fare selection, passenger information, seat selection, review/payment and confirmation.

## 15. Evaluation Requirements

Where the experience is evaluated, tasks should be based on realistic booking goals rather than isolated interface actions.

Useful evaluation questions include whether participants can identify suitable flights, understand fare differences, recognise the total price, complete passenger information, understand their selections and determine what to do after confirmation.

Any measured usability findings should be reported separately from design hypotheses. The repository should not present assumed improvements as empirical results.

## 16. Requirement Traceability

The requirements are intended to remain connected to the later design artefacts. A requirement should be visible through one or more of the following: information architecture, user flow, wireframe, component, high-fidelity screen, prototype interaction or evaluation criterion.

This makes it possible to review the final interface not only for visual quality, but also for whether it addresses the problems and objectives established earlier in the project.

## 17. Out of Scope

The following are outside the requirements of this conceptual design study: live airline inventory integration, production payment processing, operational airline-system integration, real booking issuance, backend implementation, production security architecture and commercial deployment.

## 18. Requirement Status

These requirements serve as the working design baseline for the SkyPath case study. They may be refined as the project evolves, particularly where research, prototyping or evaluation identifies a need for clarification.

## 19. Closing Note

A strong booking experience is not defined only by how attractive its final screens look. These requirements provide a way to evaluate whether the interface helps users understand information, make decisions, recover from mistakes and move through the journey with confidence.